Lambda1/100/1000 continuation | Cantera vs NN NRMSE

Median of per-IC NRMSE on the same 0 complete ICs for all displayed variants

$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

Model / same-state mean	lambda / stage	T	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}	Complete ICs
1 ANN10	**1 / soft10k**	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	17/18
1 ANN10	**1 / long**	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	18/18
2 P-only DeepONet10	**1 / soft10k**	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	18/18
2 P-only DeepONet10	**1 / long**	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	18/18
3 ANN10 + P-only DeepONet10	**1 / soft10k**	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	18/18
3 ANN10 + P-only DeepONet10	**1 / long**	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	18/18
4 T/P DeepONet10	**1 / soft10k**	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	9/18
4 T/P DeepONet10	**1 / long**	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	14/18
5 ANN10 + T/P DeepONet10	**1 / soft10k**	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	18/18
5 ANN10 + T/P DeepONet10	**1 / long**	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	18/18
6 T/P/ϕ DeepONet10	**1 / soft10k**	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	11/18
6 T/P/ϕ DeepONet10	**1 / long**	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	13/18
7 ANN10 + T/P/ϕ DeepONet10	**1 / soft10k**	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	17/18
7 ANN10 + T/P/ϕ DeepONet10	**1 / long**	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	18/18
1 ANN10	**100 / soft10k**	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	18/18
1 ANN10	**100 / long**	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	18/18
2 P-only DeepONet10	**100 / soft10k**	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	6/18
2 P-only DeepONet10	**100 / long**	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	1/18
3 ANN10 + P-only DeepONet10	**100 / soft10k**	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	18/18
3 ANN10 + P-only DeepONet10	**100 / long**	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	13/18
4 T/P DeepONet10	**100 / soft10k**	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	18/18
4 T/P DeepONet10	**100 / long**	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	3/18
5 ANN10 + T/P DeepONet10	**100 / soft10k**	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	18/18
5 ANN10 + T/P DeepONet10	**100 / long**	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	18/18
6 T/P/ϕ DeepONet10	**100 / soft10k**	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	16/18
6 T/P/ϕ DeepONet10	**100 / long**	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	4/18
7 ANN10 + T/P/ϕ DeepONet10	**100 / soft10k**	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	18/18
7 ANN10 + T/P/ϕ DeepONet10	**100 / long**	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	18/18
1 ANN10	**1000 / soft10k**	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	18/18
1 ANN10	**1000 / long**	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	18/18
2 P-only DeepONet10	**1000 / soft10k**	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	18/18
2 P-only DeepONet10	**1000 / long**	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	18/18
3 ANN10 + P-only DeepONet10	**1000 / soft10k**	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	18/18
3 ANN10 + P-only DeepONet10	**1000 / long**	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	18/18
4 T/P DeepONet10	**1000 / soft10k**	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	18/18
4 T/P DeepONet10	**1000 / long**	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	18/18
5 ANN10 + T/P DeepONet10	**1000 / soft10k**	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	18/18
5 ANN10 + T/P DeepONet10	**1000 / long**	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	18/18
6 T/P/ϕ DeepONet10	**1000 / soft10k**	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	18/18
6 T/P/ϕ DeepONet10	**1000 / long**	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	18/18
7 ANN10 + T/P/ϕ DeepONet10	**1000 / soft10k**	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	18/18
7 ANN10 + T/P/ϕ DeepONet10	**1000 / long**	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	18/18

Exact user util.py RMS normalization: equal time-sample weights; raw dimensionless ratio without x100.
—: incomplete, no full-horizon score. N/A: empty common cohort or undefined reference RMS. Constant nonzero N2 is valid.
Short10000 updates; long endpoints have different update counts. Both models in every mixed team use the same stage/lambda.
Common ICs: .